

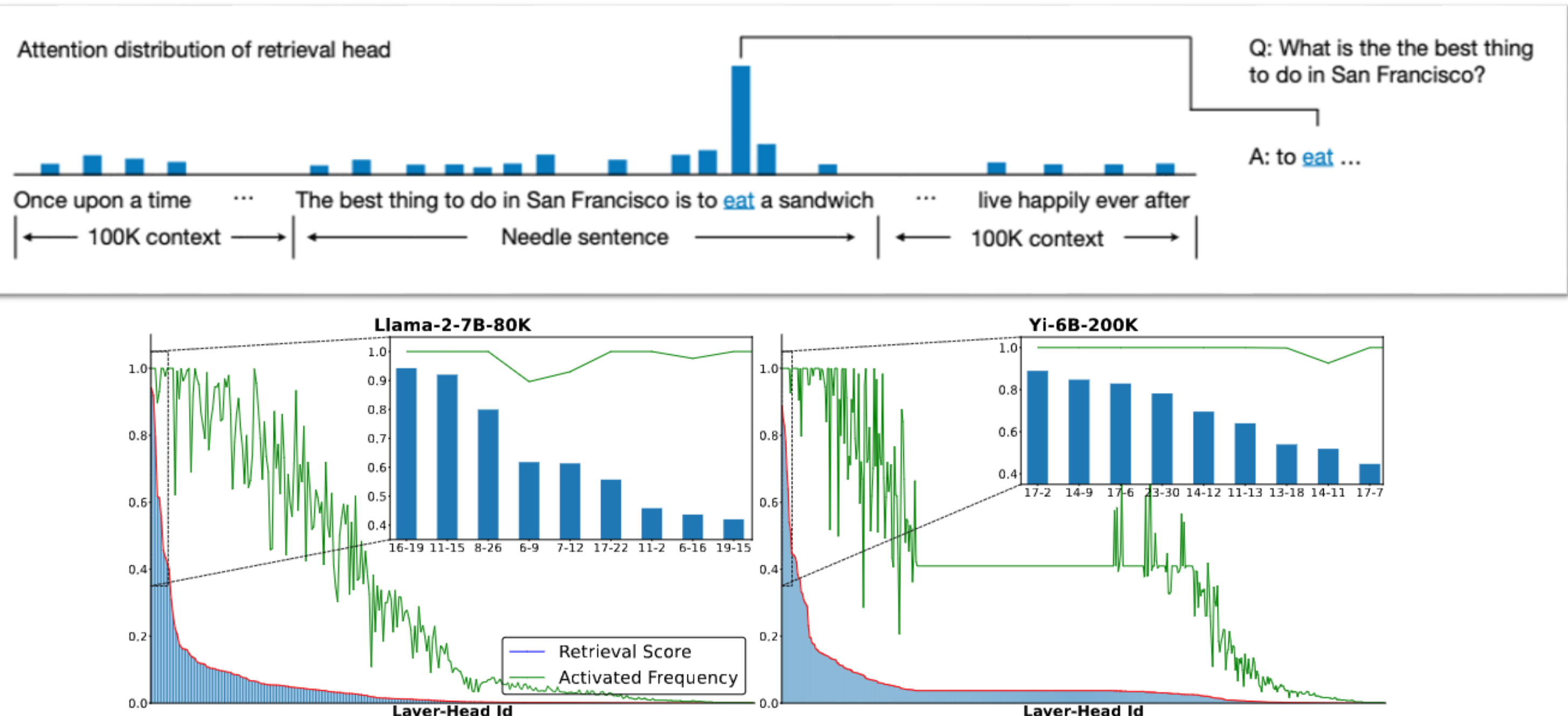
How Do Large Vision-Language Models See Text in Image? Unveiling the Distinctive Role of OCR Heads

Ingeol Baek, Hwan Chang, Sunghyun Ryu, Hwanhee Lee

Language Intelligence lab, Chung-Ang University
Department of Computer Engineering, Sejong University

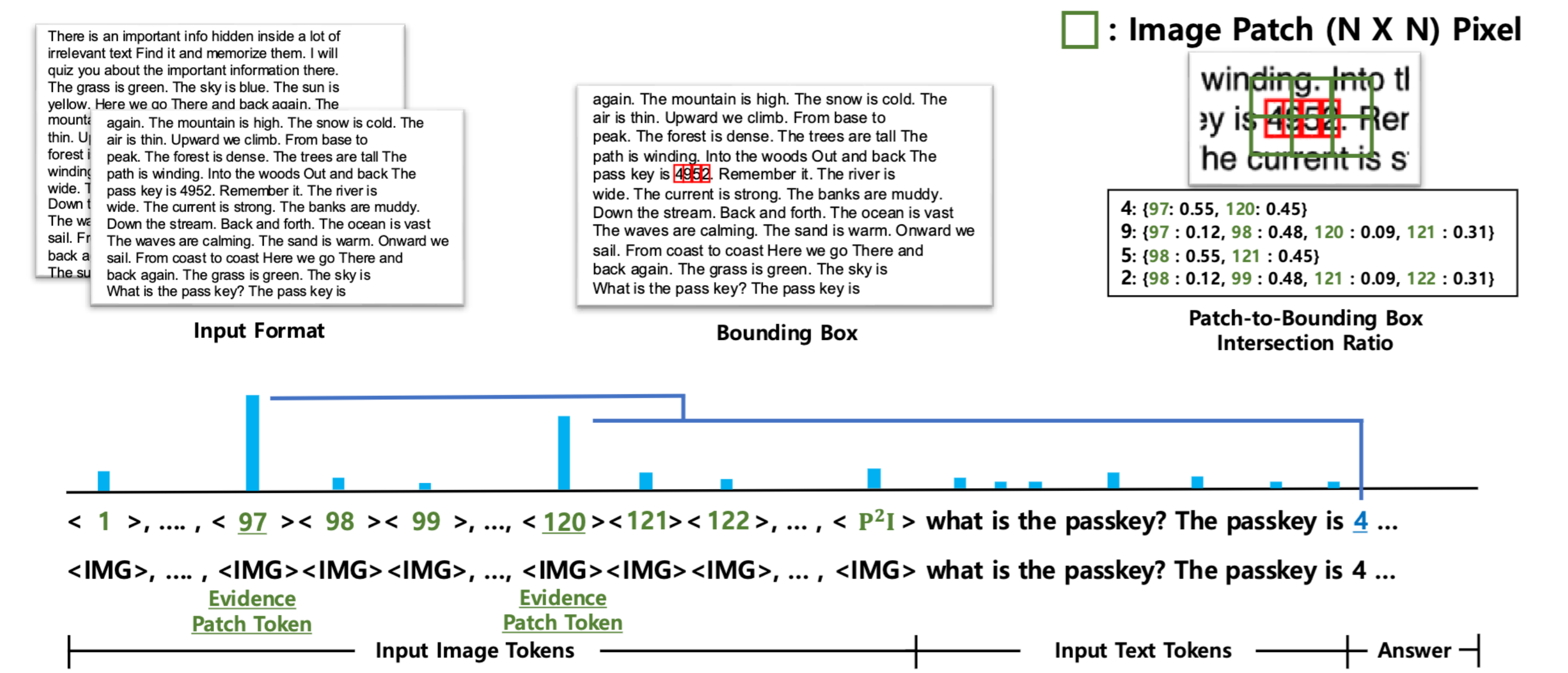


Motivation



- We hypothesize that certain attention heads perform **copy-paste** of the correct answer from the context.
- To identify them, we use a **needle-in-a-haystack (NIAH)** dataset that embeds **intermediate answers**.
- While generating the final answer, we assign that head a **Retrieval Score** if, for the gold answer tokens, the head's **maximum attention** falls on the corresponding answer tokens in the context.
- The large gap between their **Retrieval Score** and **Activated Frequency** suggests that these heads are also activated for functions **beyond retrieval**, not exclusively for copying answers.

Method



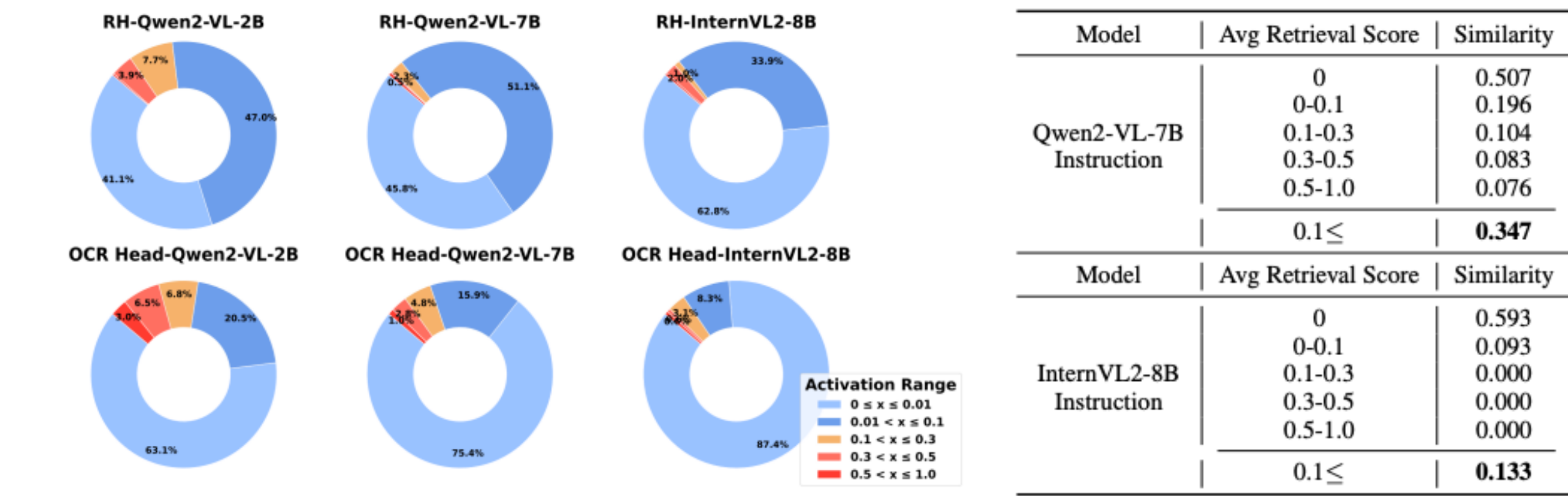
- **Step1:** Image passkey and Needle in a Haystack Dataset Construction
the original text data is converted into text images of a fixed size. Sentences are split whenever they reach a predefined string length, and a new image is generated once the limit is exceeded using a rule-based approach. For the gold answer context, bounding boxes are additionally created to mark the relevant regions.
- **Step2:** Patch Token Preprocessing
assuming a ViT patch size of 14×14 , an image of size 294×196 produces $294 / 14 \times 196 / 14 = 294$ tokens that serve as input to the LLM. When multiple images are considered, the total number of input tokens is calculated as $(w / N) \times (h / N) \times I$, where w and h denote the image width and height, N the patch size, and I the number of images.

- **Step3:** Evidence Patch Tokens
The process for creating the **Patch-to-Bounding Box Intersection Ratio** shown above involves checking how much each patch overlaps with the bounding box. For example, in the case of the gold answer digit “4”, the 97th patch token overlaps by **55%**, while the 120th patch token overlaps by **45%**. Patches with less than **10% overlap** are discarded.

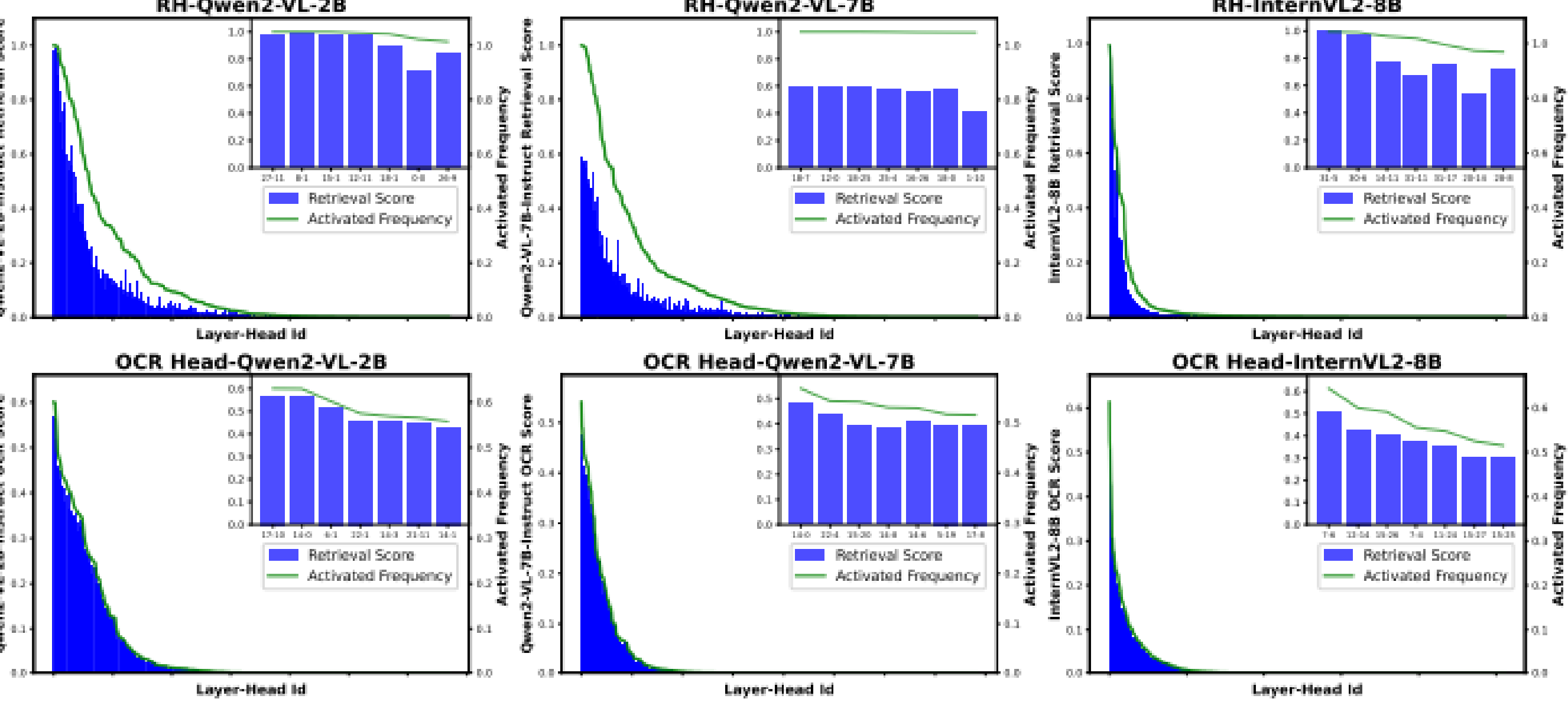
- **Step4:** OCR Score
The generated token is considered correct if its **maximum attention score** points to a token within the **Patch-to-Bounding Box Intersection Ratio**. In this case, the generated token corresponds to the **gold answer token**.

- **Step5:** Detecting OCR Head
Using **600 synthetic Image Passkey samples** and **600 Needle-in-a-Haystack samples**, we classify a head as an **OCR head** if it is activated more than **10% on average** and achieves an **OCR Score above 0.1**.

Experiment



- Prior work notes that **Retrieval Heads** are activated **sparsely**. In contrast, **OCR Heads** show **lower sparsity**, meaning they activate more frequently.
- Given their copy-paste-like behavior, similar to Retrieval Heads, we measured **head-to-head similarity** using the **Jaccard index**. The similarity of activated heads turned out to be relatively low: **0.347** in **Qwen2-VL** and **0.133** in **InternVL2**.



- **Retrieval Heads** show a large gap between **frequency** and **score**, indicating that they are **dynamically activated**. In contrast, **OCR Heads** show almost no such gap, suggesting that they are **statically activated**.

Analysis

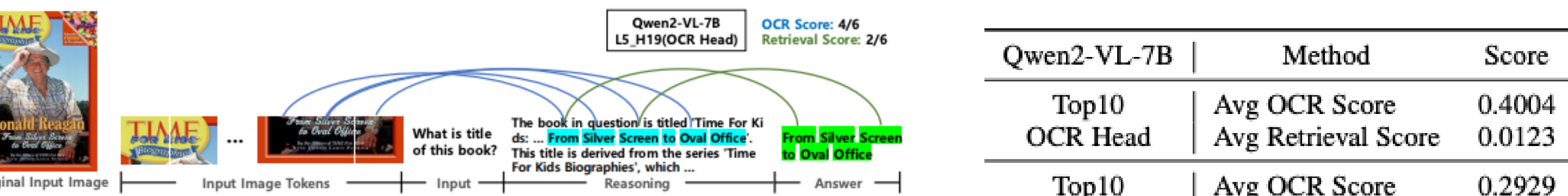
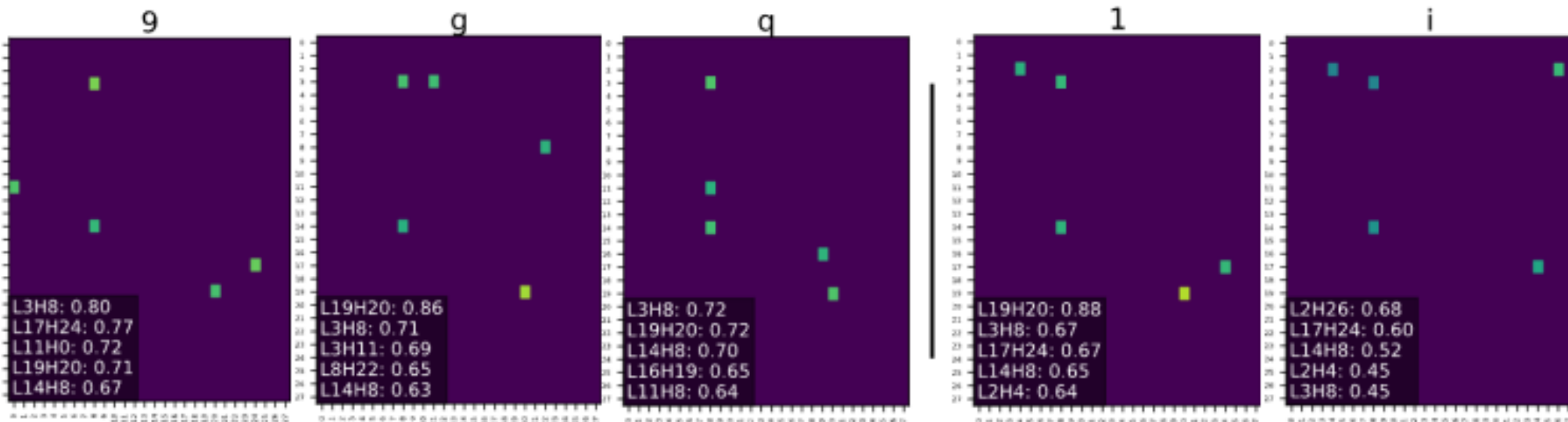


Figure 5: In the OCR-VQA task, we calculate the OCR score and Retrieval score based on the CoT prompting. L5_H19 refers to Layer 5, Head 19. The blue and green lines indicate the token positions with the highest attention.

- In the **CoT setting**, we measure both the **OCR Score** (from OCR heads) and the **Retrieval Score** (from retrieval heads), based on (i) the part where the answer is mentioned during reasoning and (ii) the part where the answer is actually generated.
- We find that in **OCR heads**, the retrieval score is almost absent, supporting the claim that they are **statically activated**.
- In contrast, in **retrieval heads**, an OCR score is also observed, indicating that they are **dynamically activated**.

			DocVQA	MP-DocVQA
Qwen2-VL-7B	Baseline		76.3	69.1
	OCR		76.6 (+0.3)	69.7 (+0.6)
	RH		76.2 (-0.1)	69.4 (+0.3)
InternVL2-8B	Baseline		39.1	35.0
	OCR		40.0 (+0.9)	35.8 (+0.8)
	RH		39.3 (+0.2)	34.7 (-0.3)

- Masking OCR heads led to a **significant performance drop** in the **VQA task**.
- Masking retrieval heads caused a **large performance decline** in the **QA task**.
- The sink token is defined as the **first token (index 0)**, and its attention weight was reduced to redistribute scores across other tokens.
- When redistribution was applied to OCR heads, we observed a **performance improvement** in the **VQA task**.



- For visually similar characters such as “i” and “1”, the same four heads (**L3H8**, **L2H4**, **L14H8**, and **L17H24**) were activated.
- For “9”, “g”, and “q”, three heads were commonly activated across these characters.